

# MODEL PAPER

## Periodic Assessment-1

### MVGR College of Engineering (Autonomous)

Program: B.Tech

Regulation: R24

Course Code: R24MSCST006

Semester: III

#### Principles of Programming Languages

(Common to CSE, CSIT, IT, CSE(DS), CSE(ICB), CSE(AIML))

Duration: 90 Minutes

Max. Marks.: 21.

Answer All Questions.

All Questions Carry EQUAL Marks.

1. Construct a program to create a 2D rectangular array and print the sum of each row. 7M  
  
(OR)
2. Define the following terms in the context of BNF: 7M  
(a) Terminal symbol (b) Non-terminal symbol (c) Production rule  
(d) Derivation (e) Start symbol
3. Develop a menu-based program using switch-case in C for a basic calculator. 7M  
  
(OR)
4. List the rules of associativity and precedence for arithmetic operators in C with examples. 7M
5. Define the following memory management terms in the context of C programming: 7M  
(a) Stack (b) Heap (c) malloc() (d) free() (e) local variable  
  
(OR)
6. List and define two types of type equivalence and give examples. 7M

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